

THE RAMANUJAN BOUND IS THE TEMPERED BOUND: THE HECKE OPERATOR ON THE BRUHAT–TITS TREE, AND A NO-GO THEOREM FOR CONDITION (C4)

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ABSTRACT. Condition (C4) of Paper X of this series asks whether the measured groupoid of the Hurwitz quaternion Bost–Connes system is non-amenable, equivalently whether its factor escapes the injectivity that erased the arithmetic in every earlier model. Paper XI reduced the question to a spectral property of a transfer operator and found the naive operator unusable, the range projections being nested rather than orthogonal; Paper XII reduced it further to a single representation of the norm-one group and warned that a spectral gap would not suffice. We settle the question.

We first build the operator the programme wanted: the Hecke operator $T_p = \sum_{k=1}^{p+1} \pi(a_k)$, the sum of the Koopman unitaries of the $p+1$ Hurwitz quaternions of reduced norm p , which is the adjacency operator of the $(p+1)$ -regular Bruhat–Tits tree of $\mathrm{GL}_2(\mathbb{Q}_p)$ transported to the Maharam extension. It involves no projections and so has none of the pathology of the transfer operator.

We then compute its spectrum exactly. For every finite set F of primes the subgroup $G_F = \mathbb{H}(\mathbb{Z}[1/F])^\times$ is discrete in $\prod_{p \in F} \mathbb{H}(\mathbb{Q}_p)^\times$ and acts *properly and freely* on a conull set, because every element of $\mathbb{H}(\mathbb{Q})^\times$ is a unit at almost every place. Hence the Koopman representation restricted to G_F is a multiple of the regular representation, and, with ε_p the central unitary of p (so that $\varepsilon_p^{-1/2} T_p$ is the self-adjoint Hecke operator),

$$\mathrm{sp}(\varepsilon_p^{-1/2} T_p \upharpoonright L_0^2) = [-2\sqrt{p}, 2\sqrt{p}], \quad \|T_p\| = 2\sqrt{p}.$$

The Ramanujan bound is attained, not merely respected. It is also worthless for the purpose: $2\sqrt{p}$ is the norm of the adjacency operator in the *regular* representation of a free group, so the gap between $2\sqrt{p}$ and the trivial eigenvalue $p+1$ is Kesten’s gap and measures the non-amenability of the *group*, not of the *action*. By Kuhn’s theorem an amenable action has tempered Koopman representation; attaining the Ramanujan bound proves temperedness, and temperedness is the signature of amenability, not an obstruction to it.

The same properness gives the verdict directly, with no spectral theory at all. The orbit relation is the increasing union over finite F of the smooth relations of the proper actions of G_F , hence hyperfinite; the groupoid is amenable and the factor is injective. **(C4) is false**, the Hurwitz factor collapses to R_∞ , R_λ or a Krieger factor like every abelian system before it, and by Paper IX it retains nothing of the arithmetic. The argument uses only that each element of the monoid is a unit at almost every place, so it closes the search of Paper X in full: conditions (C1)–(C3) are incompatible with (C4) for every order in every finite-dimensional $\mathbb{Q}$ -algebra.

1. SETTING AND STATEMENTS

$\mathbb{H}$ is the rational Hamilton quaternion algebra, ramified at 2 and ∞ ; $\mathcal{H} \subset \mathbb{H}(\mathbb{Q})$ is the Hurwitz order, $P = \mathcal{H} \setminus \{0\}$ the multiplicative Ore monoid of Paper X, Nrd the reduced norm and $G = PP^{-1} = \mathbb{H}(\mathbb{Q})^\times$ the group of right fractions [9]. For a prime p write $\mathcal{H}_p = \mathcal{H} \otimes \mathbb{Z}_p$; for odd p one has $\mathcal{H}_p \cong M_2(\mathbb{Z}_p)$ and the right ideals of $\mathcal{H}_p$ of reduced norm p are the $p+1$ neighbours of the base vertex in the Bruhat–Tits tree $\mathcal{T}_p$ of $\mathrm{GL}_2(\mathbb{Q}_p)$ [8]. The localized Bost–Connes framework in which the system sits is that of [11]; the ingredients we use from Papers IX–XII of the series are re-proved here.

Hypothesis 1.1 (The model). $\Omega = \widehat{\mathcal{H}} = \prod_p \mathcal{H}_p$ carries a KMS $_\beta$ measure ν_β of the Nica–Toeplitz algebra $\mathcal{T}(P)$, $\tilde{\Omega} = \varinjlim (\Omega, \cdot a) = \mathbb{H}(\mathbb{A}_f)$ is the dilation, G acts on $\tilde{\Omega}$ by right translation, and $\tilde{\nu}_\beta$ is the extension of ν_β , so that $\tilde{\nu}_\beta(Ea) = \text{Nrd}(a)^{-\beta} \tilde{\nu}_\beta(E)$. The Maharam extension is

$$(1.1) \quad G \curvearrowright (\tilde{\Omega} \times \mathbb{R}, \tilde{\nu}_\beta \otimes e^s ds), \quad a \cdot (x, s) = (xa, s + \beta \log \text{Nrd}(a)),$$

which preserves the displayed infinite measure. Condition (C4) of Paper X asks whether the measured groupoid $G \ltimes (\tilde{\Omega}, \tilde{\nu}_\beta)$ is non-amenable, equivalently — by Takesaki duality, as in Paper XI — whether (1.1) is non-amenable, equivalently whether $M_\beta = \pi_\varphi(\mathcal{T}(P))''$ fails to be injective.

Nothing below uses the value of β , the existence of a KMS state in any particular temperature range, or the type of M_β . Only the following is used, and it is a property of the arithmetic and not of the dynamics.

Lemma 1.2 (Finite support). *Every $a \in G = \mathbb{H}(\mathbb{Q})^\times$ lies in $\mathcal{H}_q^\times$ for all but finitely many primes q .*

Proof. Write $a = b/m$ with $b \in \mathcal{H}$ and $m \in \mathbb{Z}_{>0}$. For $q \nmid m \text{Nrd}(b)$ one has $b \in \mathcal{H}_q$ and $\text{Nrd}(b) \in \mathbb{Z}_q^\times$, so $b\bar{b} = \text{Nrd}(b)$ is a unit and $b \in \mathcal{H}_q^\times$; and $m \in \mathbb{Z}_q^\times$. Only the finitely many q dividing $m \text{Nrd}(b)$ are excluded. $\square$

For a finite set F of primes put

$$G_F = \{a \in G : a \in \mathcal{H}_q^\times \text{ for every } q \notin F\} = \mathbb{H}(\mathbb{Z}[1/F])^\times, \quad L_F = \prod_{p \in F} \mathbb{H}(\mathbb{Q}_p)^\times.$$

By Lemma 1.2, $G = \bigcup_F G_F$, the union being increasing and countable.

The three results are the following.

Theorem 1.3 (Local properness). *For every finite F , G_F is a discrete subgroup of L_F and its action on $\tilde{\Omega}$ is free and proper on a conull invariant set. Consequently the orbit relation $\mathcal{R}_F$ is smooth, and the Koopman representation of the Maharam extension restricted to G_F is unitarily equivalent to a multiple of the regular representation λ_{G_F} .*

Theorem 1.4 (The spectrum of the Hecke operator). *Let p be an odd prime, $a_1, \dots, a_{p+1} \in \mathcal{H}$ generators of the $p+1$ right ideals of reduced norm p , and $T_p = \sum_{k=1}^{p+1} \pi(a_k)$ the associated Hecke operator on $L^2(\tilde{\Omega} \times \mathbb{R})$ of Definition 2.1. Then T_p is the adjacency operator of $\mathcal{T}_p$ in the regular representation up to the central unitary $\varepsilon_p = \pi(p)$, and on the orthogonal complement L_0^2 of the invariant vectors the self-adjoint operator $\varepsilon_p^{-1/2} T_p$ of Lemma 2.2(1) satisfies*

$$\text{sp}(\varepsilon_p^{-1/2} T_p \upharpoonright L_0^2) = [-2\sqrt{p}, 2\sqrt{p}], \quad \|T_p \upharpoonright L_0^2\| = 2\sqrt{p}.$$

(T_p itself is normal but not self-adjoint, $T_p^ = \varepsilon_p^{-1} T_p$; its spectrum is the disc swept by the central phases, and the invariant statement is the one displayed.) The Ramanujan bound is attained, and the trivial eigenvalue $p+1$ is absent from L_0^2 .*

Theorem 1.5 (No-go for (C4)). *The orbit relation $\mathcal{R}_G = \bigcup_F \mathcal{R}_F$ is hyperfinite. Hence the measured groupoid of Hypothesis 1.1 is amenable, (C4) is false, and M_β is injective: it is R_∞ , a Powers factor R_λ , a Krieger factor, or type I. By Paper IX it therefore retains no arithmetic invariant of the system.*

Theorem 1.5 does not use Theorem 1.4; the spectral computation is included because it identifies precisely what the programme of Paper XI was measuring, and why the measurement could not have decided the question. Section 6 extends the no-go to every candidate monoid admissible under (C1)–(C3).

2. THE HECKE OPERATOR

Paper XI worked with the transfer operator $\mathcal{M}_p(x) = \frac{1}{p+1} \sum_k v_{a_k}^* x v_{a_k}$ built from the isometries of $\mathcal{T}(P)$. Its defect is structural: the range projections $e_{a_k} = v_{a_k} v_{a_k}^*$ are nested, $e_{a_j} e_{a_k} = e_p \neq 0$, so $\mathcal{M}_p$ is not a Markov operator of the tree and forcing orthogonality collapses the algebra. The remedy is to work on the Maharam extension, where the generators act by *unitaries* and no projection occurs.

Definition 2.1. Let π be the Koopman representation of (1.1),

$$(\pi(a)f)(x, s) = f(xa, s + \beta \log \text{Nrd}(a)), \quad a \in G,$$

a unitary representation of G on $L^2(\tilde{\Omega} \times \mathbb{R}, \tilde{\nu}_\beta \otimes e^s ds)$ because the Maharam extension preserves the measure. For odd p fix representatives $a_1, \dots, a_{p+1} \in \mathcal{H}$ of the $p+1$ right ideals of reduced norm p and set

$$T_p = \sum_{k=1}^{p+1} \pi(a_k).$$

Lemma 2.2. With $\varepsilon_p = \pi(p)$ the unitary of the central element p :

- (1) the set $\{a_k\}$ may be chosen closed under $a \mapsto \bar{a}$ modulo right units, and then $T_p^* = \varepsilon_p^{-1} T_p$, so $\varepsilon_p^{-1/2} T_p$ is self-adjoint for either square root;
- (2) $T_p^2 = T_{p^2} + (p+1)\varepsilon_p$, where T_{p^2} is the sum over the $p(p+1)$ right ideals of reduced norm p^2 not containing $p\mathcal{H}$;
- (3) under the identification of Theorem 1.3, T_p is the adjacency operator of the $(p+1)$ -regular tree $\mathcal{T}_p$ acting on ℓ^2 of its vertex set.

Proof. (1) $a_k \bar{a}_k = p$, so $\pi(\bar{a}_k) = \varepsilon_p \pi(a_k)^{-1} = \varepsilon_p \pi(a_k)^*$; summing over a self-conjugate set of representatives gives the identity. (2) Expanding $T_p^2 = \sum_{j,k} \pi(a_j a_k)$, the products $a_j a_k$ of reduced norm p^2 either generate an ideal not containing $p\mathcal{H}$, and each such ideal arises exactly once, or satisfy $a_j a_k \in p\mathcal{H}$, which happens for exactly one k for each j : $a_j a_k \in p\mathcal{H}$ forces $a_k \mathcal{H} = \bar{a}_j \mathcal{H}$, and with the self-conjugate choice of (1) then $a_k = \bar{a}_j$ exactly, so the product is $a_j \bar{a}_j = p$ and contributes ε_p exactly; there are $p+1$ such pairs. (3) The vertices of $\mathcal{T}_p$ at distance n from the base vertex are the right ideals of $\mathcal{H}_p$ of reduced norm p^n that are not divisible by p beyond the necessary amount, and $a \mapsto$ (the ideal it generates) intertwines right multiplication by the a_k with the adjacency of $\mathcal{T}_p$; the identification of the Hilbert space is Corollary 3.1. (Vertices are ideal classes up to homothety, i.e. after trivializing the central $\langle p \rangle$; on the Koopman space itself the central unitary ε_p survives, which is why the self-adjoint operator is $\varepsilon_p^{-1/2} T_p$.) $\square$

Remark 2.3. By Lubotzky–Phillips–Sarnak the group generated by $a_1, \dots, a_{p+1}$ modulo the centre and the 24 units is free of rank $(p+1)/2$ and acts simply transitively on the vertices of $\mathcal{T}_p$; the finite quotients by congruence subgroups are the Ramanujan graphs $X^{p,q}$. This is the algebraic rigidity the present programme hoped to exploit.

3. LOCAL PROPERNESS AND THE REGULAR REPRESENTATION

Proof of Theorem 1.3. Discreteness. Let $a_n \in G_F$ with $a_n \rightarrow 1$ in L_F . Then $\text{Nrd}(a_n) \rightarrow 1$ in $\prod_{p \in F} \mathbb{Q}_p^\times$, and $\text{Nrd}(a_n) \in \mathbb{Z}[1/F]^\times = \{\pm \prod_{p \in F} p^{e_p}\}$, so for large n all exponents vanish and $\text{Nrd}(a_n) = 1$. Thus a_n lies in the norm-one group $\mathbb{H}(\mathbb{Z}[1/F])^1$, which is a lattice in $\prod_{p \in F} \mathbb{H}(\mathbb{Q}_p)^1$ because $\mathbb{H}$ is definite and $\mathbb{H}(\mathbb{R})^1 = SU(2)$ is compact; in particular it is discrete, so $a_n = 1$ for large n .

Properness and freeness. Write $\tilde{\Omega} = X_F \times Y_F$ with $X_F = \prod_{p \in F} \mathbb{H}(\mathbb{Q}_p)$ and Y_F the remaining coordinates. The locus where the component at p has zero reduced norm is $\tilde{\nu}_\beta$ -null: an element of

$\mathcal{H}_p$ of zero reduced norm lies, for every n , in one of the right ideals of reduced norm p^n , of which there are $\sigma(p^n) = 1 + p + \cdots + p^n$, each carried by the scaling relation to measure $p^{-n\beta}$ times that of the base cylinder (an element of reduced norm p^n is a unit at every other place, by Lemma 1.2, so the cylinder is an honest translate); the bound $\sigma(p^n)p^{-n\beta} \rightarrow 0$ holds for $\beta > 1$, in particular throughout the Gibbs range $\beta > 2$ of Paper X. So $X_F^\times = \prod_{p \in F} \mathbb{H}(\mathbb{Q}_p)^\times$ is conull and invariant. By Lemma 1.2 and the definition of G_F , an element $a \in G_F$ acts on Y_F through the compact group $\prod_{q \notin F} \mathcal{H}_q^\times$ and on $X_F^\times$ by right translation by the discrete subgroup $G_F \leq L_F$. Right translation of a locally compact group by a discrete subgroup is free and proper, and properness in one factor of a product is properness; so the action on $X_F^\times \times Y_F$ is free and proper.

Smoothness and the Koopman representation. A free proper action of a countable group admits a measurable fundamental domain D , so $\tilde{\Omega} \times \mathbb{R} = \bigsqcup_{a \in G_F} (D \cdot a)$ up to null sets and $\mathcal{R}_F$ is smooth. Trivializing the Radon–Nikodym derivative by half-densities gives a unitary $L^2(\tilde{\Omega} \times \mathbb{R}) \cong L^2(D) \otimes \ell^2(G_F)$ intertwining $\pi \upharpoonright G_F$ with $1 \otimes \lambda_{G_F}$. $\square$

Corollary 3.1. $\pi \upharpoonright G_F \cong \dim L^2(D) \cdot \lambda_{G_F}$ for every finite F . In particular $\pi \upharpoonright G_F$ is tempered, and for every ξ in the group algebra $\|\pi(\xi)\| = \|\lambda_{G_F}(\xi)\|$.

4. THE SPECTRUM, AND WHAT IT DOES NOT PROVE

Proof of Theorem 1.4. Take $F = \{p\}$, so $T_p = \pi(\xi)$ with $\xi = \sum_k a_k$ in the group algebra of $G_{\{p\}}$. By Corollary 3.1, $\|T_p\| = \|\lambda(\xi)\|$ and more precisely $\text{sp}(T_p) = \text{sp}_{L(G_{\{p\}})}(\xi)$, the spectrum in the group von Neumann algebra. Modulo the centre and the 24 units, the a_k generate a group Γ acting simply transitively on the vertices of $\mathcal{T}_p$ — free of rank $(p+1)/2$ when $p \equiv 1 \pmod{4}$ [7], a free product of cyclic groups in general — and ξ maps to the sum of the $p+1$ generating translations, whose image under λ_Γ is the adjacency operator of the Cayley graph of Γ , which is $\mathcal{T}_p$ [8]. To pass from $\ell^2(G_{\{p\}})$ to ℓ^2 of the vertex set, decompose over the characters of the central subgroup generated by p and the finite unit group: in each fibre T_p acts as a phase-twisted tree adjacency operator, and on a tree every phase twist is gauge-trivial, so the fibre operator is unitarily a phase times the tree adjacency, and the self-adjoint $\varepsilon_p^{-1/2} T_p$ has in every fibre the spectrum of the tree adjacency itself. By Kesten’s theorem [5] the spectrum of the adjacency operator of the $(p+1)$ -regular tree on ℓ^2 of its vertices is exactly $[-2\sqrt{p}, 2\sqrt{p}]$, with norm $2\sqrt{p}$ and no eigenvalues. Invariant vectors would carry the eigenvalue $p+1$ and do not lie in ℓ^2 , so they are absent from L_0^2 . $\square$

Corollary 4.1 (There is a gap, and it is Kesten’s). $\|T_p \upharpoonright L_0^2\| = 2\sqrt{p} < p+1$, a gap of relative size $1 - 2\sqrt{p}/(p+1)$:

p	3	5	7	11	13	101
$2\sqrt{p}$	3.464	4.472	5.292	6.633	7.211	20.10
$p+1$	4	6	8	12	14	102
gap	0.134	0.255	0.339	0.447	0.485	0.803

This gap is present in the regular representation of Γ itself, and its existence is equivalent, by Kesten’s criterion [5], to the non-amenability of the free group Γ . It is therefore a statement about the group G , not about the action of G on $\tilde{\Omega}$.

Theorem 4.2 (The proposed equivalence is vacuous). *The following are equivalent, and none of them implies non-amenability of the action:*

- (1) $\|T_p \upharpoonright L_0^2\| = 2\sqrt{p}$, the Ramanujan bound, for every odd p ;
- (2) $\pi \upharpoonright G_F$ is weakly contained in λ_{G_F} for every finite F , i.e. the Koopman representation is tempered;
- (3) the free group Γ is non-amenable.

Indeed (1) and (2) hold, by Theorem 1.4 and Corollary 3.1, while the action is amenable by Theorem 1.5.

Proof. (1) $\Leftrightarrow$ (3) is Kesten's criterion [5]. (2) $\Rightarrow$ (1) because temperedness gives $\|\pi(\xi)\| \leq \|\lambda(\xi)\|$ and the reverse inequality is Corollary 3.1; (1) $\Rightarrow$ (2) for the algebra generated by the a_k , and the general case is Corollary 3.1 again. The last assertion is the content of the paper. $\square$

Remark 4.3 (The orientation of the argument). Kuhn's theorem [6], quoted in Paper XII, says that an amenable action has tempered Koopman representation. Establishing the Ramanujan bound therefore establishes temperedness, which is a *necessary condition for amenability*, not an obstruction to it. To prove (C4) by spectral means one would have to exhibit spectrum of T_p outside $[-2\sqrt{p}, 2\sqrt{p}]$ on L_0^2 — that is, a *violation* of the Ramanujan property, the exact opposite of what the arithmetic of the Hurwitz order provides. The bound $2\sqrt{p}$ is not a strong constraint that the programme could exploit; it is the Plancherel measure of $\mathrm{PGL}_2(\mathbb{Q}_p)$ announcing that there is nothing beyond the regular representation to find.

5. THE NO-GO THEOREM

Proof of Theorem 1.5. By Lemma 1.2 the sets $\mathcal{R}_F$, F finite, form an increasing sequence of subrelations of $\mathcal{R}_G$ with union $\mathcal{R}_G$. By Theorem 1.3 each $\mathcal{R}_F$ is smooth, hence hyperfinite. An increasing union of hyperfinite equivalence relations is hyperfinite [3, 2], so $\mathcal{R}_G$ is hyperfinite, hence amenable [10]. Amenability of the orbit relation is amenability of the measured groupoid, since the isotropy is trivial on the conull set of Theorem 1.3 and the relation is the one attached to the groupoid by [4]; amenability of the groupoid is equivalent to injectivity of the associated von Neumann algebra by [2, 1]. The classification of injective factors then leaves only R_∞ , R_λ , a Krieger factor and the type I case, and the reconstruction failure is Paper IX. $\square$

Remark 5.1 (Why no spectral input was needed). The proof does not see the tree, the Ramanujan property or the non-amenability of G . It uses one arithmetic fact, Lemma 1.2: an element of G has a reduced norm with finitely many prime factors, so it acts at finitely many places at a time. The group never acts *all at once*. Non-amenability of G is a statement about the group acting on itself; the groupoid only ever sees the finite-level subgroups G_F , each of which acts properly.

Remark 5.2 (The F_2 example was the mechanism). Paper XII observed, as a warning about Zimmer's theorem, that the translation action of F_2 on itself preserves counting measure, is essentially free, and is nonetheless amenable, being proper. Theorem 1.3 shows that this is not an analogy: the Hurwitz action *is* an increasing union of translation actions of discrete groups on locally compact groups. The counterexample and the system coincide.

6. THE SEARCH OF PAPER X IS EMPTY

Theorem 6.1 (General no-go). *Let A be a finite-dimensional $\mathbb{Q}$ -algebra with reduced norm Nrd , $\Lambda \subset A$ an order, $P \subseteq \Lambda \setminus \{0\}$ a right Ore, right LCM monoid with $G = PP^{-1} \leq A^\times$ and $N = |\mathrm{Nrd}|$ the norm of Paper X. Suppose G acts by right translation on the finite-adelic space $\tilde{\Omega} = A(\mathbb{A}_f)$ with a quasi-invariant measure scaling by $N^{-\beta}$, and that G is discrete in $\prod_{p \in F} A(\mathbb{Q}_p)^\times$ for every finite F — automatic when the orders of A have finite unit groups, as for definite quaternion algebras over $\mathbb{Q}$; for orders with infinite unit groups it can fail, already for a real quadratic field, where the closure of the unit group in the p -adic factors is the Leopoldt object of Papers XXII–XXIII, and the hypothesis must then be checked. Then the orbit relation is hyperfinite and the associated factor is injective, for every β .*

Proof. Lemma 1.2 holds verbatim: $a \in G$ has $\mathrm{Nrd}(a)$ a rational number, so a is a unit of $\Lambda \otimes \mathbb{Z}_q$ for all but the finitely many q dividing its numerator and denominator. The proof of Theorem 1.3 then applies with $\mathbb{H}$ replaced by A , and the proof of Theorem 1.5 follows. $\square$

Corollary 6.2. *Conditions (C1)–(C3) of Paper X are incompatible with (C4). Any monoid satisfying (C1) sits inside an order in a finite-dimensional $\mathbb{Q}$ -algebra, by (C2) and the finiteness of the norm; (C3) forces the unit group $P^\times$ to be finite (Paper X, its Remark 1.5: an infinite unit group makes ζ_P diverge identically), so the norm-one group of the order is finite, the archimedean norm-one group is compact, and the discreteness hypothesis of Theorem 6.1 holds; and every element of the monoid is a unit at almost every place; so the resulting groupoid is amenable and the factor is injective. The search for a non-injective arithmetic Bost–Connes factor cannot succeed by enlarging the index monoid.*

Remark 6.3 (What would be required). Theorem 6.1 isolates the obstruction exactly: the finite support of the norm. An index monoid escaping it would have to contain elements that are non-units at infinitely many places, hence of infinite norm, contradicting the multiplicativity of N into $[2, \infty)$ required by (C1). Within the framework of Paper X, therefore, no escape exists. Enlargements outside it — an index set that is not of arithmetic type, or a coefficient algebra that is not a function algebra — are excluded by Paper II and Paper XXVII respectively.

7. VERDICT

Statement	Status	Where
T_p is the tree adjacency operator	true	Lem. 2.2
G_F acts properly and freely	true	Thm. 1.3
$\pi \upharpoonright G_F \cong \infty \cdot \lambda_{G_F}$	true	Cor. 3.1
$\text{sp}(\varepsilon_p^{-1/2} T_p \upharpoonright L_0^2) = [-2\sqrt{p}, 2\sqrt{p}]$	true, attained	Thm. 1.4
a spectral gap exists	true, and it is Kesten’s	Cor. 4.1
the gap implies (C4)	false	Thm. 4.2
the orbit relation is hyperfinite	true	Thm. 1.5
(C4)	false	Thm. 1.5
some other monoid satisfies (C4)	false	Thm. 6.1

The Hurwitz system has everything the programme asked for. Its group of fractions is non-amenable; it carries free subgroups of every finite rank; it generates the Bruhat–Tits trees whose congruence quotients are the Ramanujan graphs; the Hecke operator on the Maharam extension is exactly the tree adjacency operator, with no nested projections and no Cuntz obstruction; and its spectrum attains the Ramanujan bound. None of it matters. The bound $2\sqrt{p}$ is the norm of the same operator in the regular representation, so attaining it certifies that the Koopman representation is regular — and a regular Koopman representation is what a free proper action has. The rigidity is real and it belongs to the group; the action never inherits it, because an element of an arithmetic monoid is a unit at almost every place and the group acts only finitely many places at a time.

The Bost–Connes phase transition therefore survives every enlargement of the index monoid compatible with its own axioms, and the factor does not. The arithmetic lives in the C^* -algebra with its flow, in the KMS simplex, and in the obstruction group; the von Neumann algebra is, and remains, hyperfinite.

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